Inverse-Dynamics-Critic-Guided Inference-Time Scaling for Humanoid Motion Generation

CS 348K Project

May 2026

Abstract

Kinematic humanoid motion generators such as MotionBricks can synthesize diverse Unitree G1 reference trajectories, but these references are not guaranteed to be dynamically executable. We study a test-time screening method with no generator or controller retraining: sample K MotionBricks candidates, score each with a MuJoCo inverse-dynamics heuristic critic, and select the lowest-risk clip. On a 39-identity K sweep, mean heuristic risk decreases from 38.90 at $K = 1$ to 15.93 at $K = 8$, and heuristic-accepted clips increase from 5/39 to 26/39. A larger K=1/K=8 extension over 105 paired motion identities reproduces the effect: mean risk decreases from 35.32 to 13.58, heuristic-accepted clips increase from 25/105 to 78/105, and 81/105 identities improve. A 105-row mode-seed audit over 15 unique local mode prompts shows K=8 broadly preserves trajectory/style proxies (mean 0.568 vs 0.583 for K=1), while a contact-quality audit improves artifact score from 0.274 to 0.248. The method is strongest on upright locomotion and expressive walking styles, weaker on crawling, and harmful for already stable idle clips. Controller experiments are negative controls: neither weak PD nor torque-limited computed-torque tracking validates per-clip critic scores. We also run the released SONIC G1 tracking-policy encoder/decoder in an approximate MuJoCo harness. After fixing the harness, K=8 is roughly neutral under learned-policy tracking on 105 K=1/K=8 pairs: mean survival changes from 2.005s to 2.054s and RMSE from 0.339 to 0.334 rad, with 98/105 falls in both groups and 0% torque saturation. A controller-aware oracle over the two available candidates improves survival to 2.299s and RMSE to 0.310 rad, while a four-way oracle over K=1/4/8/16 on the smaller 39-identity sweep improves survival from 2.134s to 2.833s ($p = 1.9\times10^{-6}$). Therefore, the contribution is a transparent inverse-dynamics screening and ranking study plus evidence that controller-in-the-loop selection or planner-policy co-training is needed, not a claim of real-robot feasibility. Because selection and evaluation use related heuristic scores, the reported gains should be interpreted as successful risk reranking under this critic, not independent physical validation.

1 Introduction

MotionBricks demonstrates real-time kinematic whole-body motion generation for animation and humanoid robotics, reporting high throughput and broad motion coverage [9]. The current public G1 preview used here exposes a mode/control interface rather than a full arbitrary text endpoint. Its limitation section also points to a central robotics gap: kinematic planners can output motions that violate hardware or contact constraints. Recent Unitree G1 pipelines address this by coupling kinematic motion sources to low-level controllers or physical plausibility optimization [1, 10]. This project investigates a smaller but practical intervention: can we reduce a fixed kinematic generator's inverse-dynamics heuristic risk at test time using an explicit dynamics-derived ranking signal?

We treat MotionBricks as a sampler. The first candidate uses the default deterministic generation path. Additional candidates enable stochastic Gumbel pose-token sampling and offset the reference seed. Each candidate is a qpos sequence. A MuJoCo inverse-dynamics critic estimates required torques, root wrench, velocities, accelerations, and jerk under exact replay, then assigns a heuristic risk score. Best-of-K selection keeps the candidate with lowest critic demand.

The method is intentionally simple. Its value is not novelty in inverse dynamics itself, but a measured characterization of inference-time scaling for humanoid kinematic references, including failures and negative controls.

2 Related Work

Kinematic humanoid generation. MotionBricks uses a modular latent generative backbone and smart primitives for real-time motion generation, with a public G1 demo and MotionBricks-powered motion generation in NVIDIA's GR00T Whole-Body Control stack [9]. CLAW similarly highlights that text-to-motion generators produce kinematic outputs that need physical grounding, and tracks references in MuJoCo with a low-level controller [1].

Physics-based humanoid tracking. DeepMimic and AMP established reinforcement-learning approaches for physically simulated character imitation and stylized control [6, 7]. PHC scales humanoid imitation and recovery, including noisy references from video or language-generated motion [5]. These works motivate our validation gap: a true feasibility claim should evaluate references under a strong learned tracking policy, not only an inverse-dynamics score.

Physical plausibility optimization. RoboForge couples text-guided generation and physical plausibility optimization, using bidirectional optimization between generator and controller [10]. Our study is complementary and narrower: it does not retrain the generator or controller, but asks how much quality can be recovered through candidate selection alone.

Inference-time scaling. Best-of-N sampling and verifier-guided selection are standard in language model reasoning [2, 4, 8]. We translate this idea to humanoid reference generation, replacing the reward model with an inverse-dynamics critic.

Text-motion evaluation. Standard text-to-motion benchmarks such as HumanML3D report learned text-motion retrieval and distributional metrics: R-Precision, matching score, FID, diversity, and multimodality [3, 11]. The public MotionBricks G1 preview used here exposes a mode/control interface rather than a full arbitrary text endpoint, so our prompt-following audit uses weaker task-proxy metrics and should not be compared to HumanML3D R-Precision.

3 Method

The robot model is the Unitree G1 MuJoCo model with 36-dimensional qpos: a floating root plus 29 actuated joints. Given a clip $q_{0:T}$, velocities are computed with quaternion-safe MuJoCo differencing and accelerations by finite difference. We call `mj_inverse` for each transition.

The final method is not fine-tuning MotionBricks and is not motion retargeting: it is candidate reranking. We do train neural surrogate critics in side experiments to imitate the heuristic score, but those models do not update the MotionBricks generator and are not used as the final selected result because clip-level neural selection underperforms the heuristic.

The scalar heuristic is

$$\begin{aligned} r = {} & 35e(\tau_{95}/\tau_{\max}, 1.0) + 20e(F_{95}^{root}, 10000) \\ & + 15e(M_{95}^{root}, 2000) + 10e(\omega_{95}, 10) \\ & + 10e(\alpha_{95}, 120) + 10e(jerk_{mean}, 1200), \end{aligned}$$

where $e(v, t) = \log(1 + \max(v - t, 0)/t)$. Action labels are implementation labels, not safety labels: critic-accept if risk ≤ 25 and p95 torque/limit ≤ 1.25; critic-review if risk ≤ 80 and torque/limit ≤ 5 but not critic-accepted; critic-reject otherwise.

Whole-clip best-of-K generates K complete clips and selects the lowest online segment-critic score, then evaluates the selected clip with the full critic. A 10-batch candidate audit found segment/full winner agreement in all audited batches, but this remains a finite sanity check. The main results therefore do not constitute an oracle best-of-K upper bound; they measure the implemented segment-critic selector.

4 Experiments

Main K sweep. The main benchmark evaluates 13 styles, 3 seeds, and $K \in \{1, 4, 8, 16\}$, producing 156 selected clips. Table 1 summarizes the result.

Table 1: Main 39-identity K sweep.

K	Mean risk	Critic-accept	Critic-reject
1	38.90	5/39	6/39
4	23.24	20/39	5/39
8	15.93	26/39	3/39
16	13.61	31/39	3/39

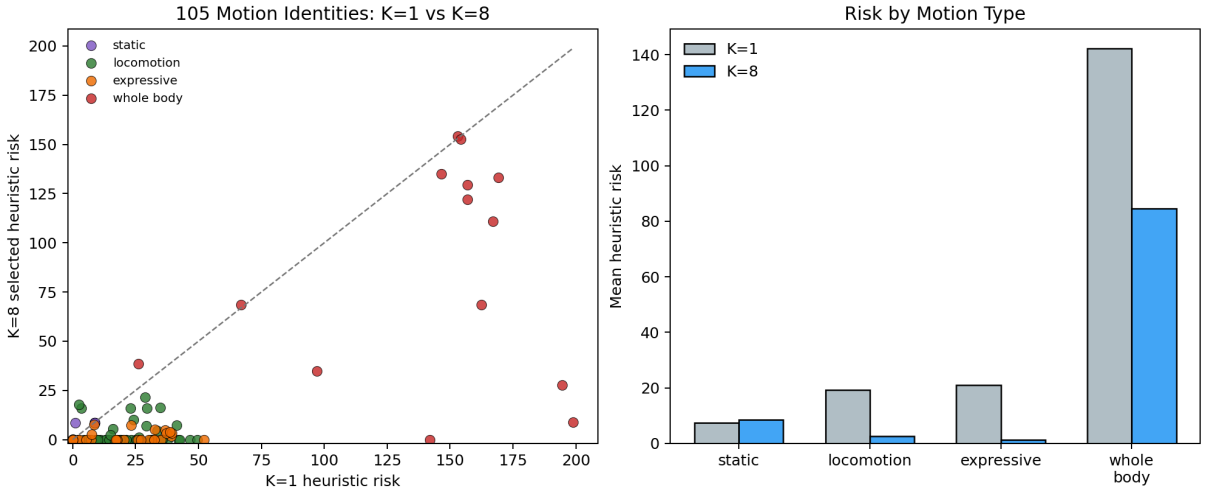

Figure 1: Extended 105-row K=1/K=8 audit. Best-of-K substantially reduces the implemented inverse-dynamics critic risk across upright locomotion and expressive modes, while whole-body crawling remains high demand.

105-identity extension. To reduce dependence on the 39-identity sweep, we evaluate all 15 exposed G1 demo modes with 7 seeds each for $K = 1$ and $K = 8$. Table 2 shows that the effect persists at larger local scale.

Table 2: Extended 105-identity K=1/K=8 comparison.

Setting	Mean risk	Median risk	Critic-accept	Critic-review	Critic-reject
K=1	35.32	21.28	25	67	13
K=8	13.58	0.00	78	18	9

The paired aggregate reduction is 61.55%, with 81/105 identities improved. By type, aggregate reductions are 86.88% for locomotion, 93.26% for expressive motions, 40.52% for whole-body crawling, and -14.38% for static idle. Static regression is expected: stochastic resampling can hurt clips that are already low demand.

Prompt/control preservation. The local preview provides discrete G1 demo modes, so the current audit is a 105-row mode-seed suite, not 100 different behaviors and not a free-form text-to-motion benchmark. It contains 15 unique local mode prompts evaluated over 7 seeds each. Each row specifies a natural-language description, category, target speed, target direction, and expected tags. We evaluate speed, direction, staticness, low-posture behavior, and upper-body expressiveness proxies.

The mean paired alignment delta is -0.015; 56/105 pairs stay within -0.05 alignment of K=1 and 72/105 maintain displacement ratio in [0.5,1.5]. The correct interpretation is that best-of-K mostly preserves local mode proxies, not that it improves language alignment.

Contact-quality audit. We also evaluate self-contact, non-foot floor contact, foot speed while contact persists, and a simple support-proxy diagnostic. The contact artifact score improves from 0.274 at K=1 to 0.248 at K=8. Whole-body crawling improves more substantially, from 0.654 to 0.489, but remains visibly problematic: non-foot

Table 3: Prompt/control proxy preservation on the 105-row mode-seed audit.

Setting	Mean align.	Median align.	Speed err.	Displacement
K=1	0.583	0.544	0.266 m/s	2.416 m
K=8	0.568	0.562	0.245 m/s	2.356 m

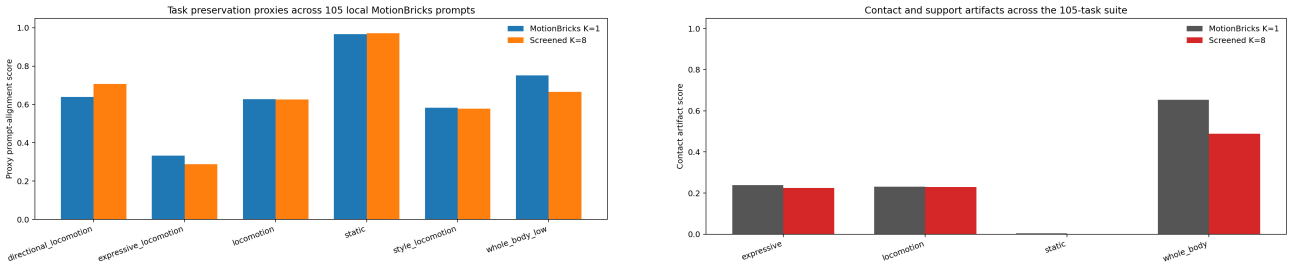

Figure 2: Damage audits. Left: prompt/control proxy scores are roughly preserved but not improved by K=8. Right: contact artifacts improve modestly overall and more on crawling, but crawling remains problematic.

floor contact is still 34.7% at K=8. These metrics help explain visible risk in videos but are not controller success metrics.

Boundary audit summary. The main K=8 result lowers critic risk by 61.55% and raises critic-accept labels from 25/105 to 78/105. The prompt/control audit is weaker: mean proxy alignment changes from 0.583 to 0.568, so this is a damage audit rather than an alignment improvement. The contact audit improves from 0.274 to 0.248 artifact score, but it is post-hoc and not part of selection. The controller audit is negative: all computed-torque rollouts fall at frame 15.

Learned-policy audit. To test whether lower inverse-dynamics risk transfers to a learned controller, we build the public SONIC G1 deployment stack and run its released encoder/decoder ONNX weights in a MuJoCo closed-loop harness. The native binary loads our exported references and TensorRT-converts the policy stack, but waits for live Unitree `LowState` DDS messages; for batch evaluation we therefore reconstruct the observation path in Python and label it an approximate harness.

The corrected result is more nuanced than the earlier buggy run. On 105 paired K=1/K=8 references, 98/105 rollouts fall in each group, while idle references survive the full 5s window. Mean survival changes from 2.005s for K=1 to 2.054s for inverse-dynamics-selected K=8, and mean joint RMSE changes from 0.339 to 0.334 rad. These differences are not significant ($p = 0.159$ for greater survival; $p = 0.457$ for lower RMSE), despite the large drop in inverse-dynamics heuristic risk.

We then use the 39-identity K sweep, which provides K=1/4/8/16 screened variants, as a limited controller-in-the-loop oracle. Selecting the variant with longest SONIC survival improves mean survival from 2.134s at K=1 to 2.833s ($p = 1.9\times10^{-6}$) and lowers RMSE from 0.328 to 0.286 rad ($p = 0.019$). The fall count remains 36/39 because idle clips survive in all settings. This supports the pivot: physical awareness should be measured with the target tracking policy in the loop, but the current candidate pool and policy interface are not sufficient for a real-robot execution claim.

Ablations and negative controls. Whole-clip K=4 outperforms per-segment K=4 at equal compute on the current setup. Gumbel sampling is the strongest diversity source, while seed diversity helps expressive motions. A neural window critic reaches high window-level agreement with heuristic labels, but neural full-clip selection is worse than heuristic selection, so it is not used for final results. Retraining a full-clip critic on all 210 extended clips improves validation Spearman correlation to 0.747. A larger 3.38M-parameter residual temporal CNN trained for 800 epochs on 497 unique labeled clips improves correlation to 0.800, still below the 0.85 target for replacing the heuristic selector.

Most importantly, controller validation is currently negative. This is the main negative result of the study. The weak PD controller has insignificant per-clip correlation with risk (Spearman $\rho = -0.09$). A stronger torque-limited computed-torque tracker also fails uniformly at frame 15 and does not improve K=8 tracking RMSE. The

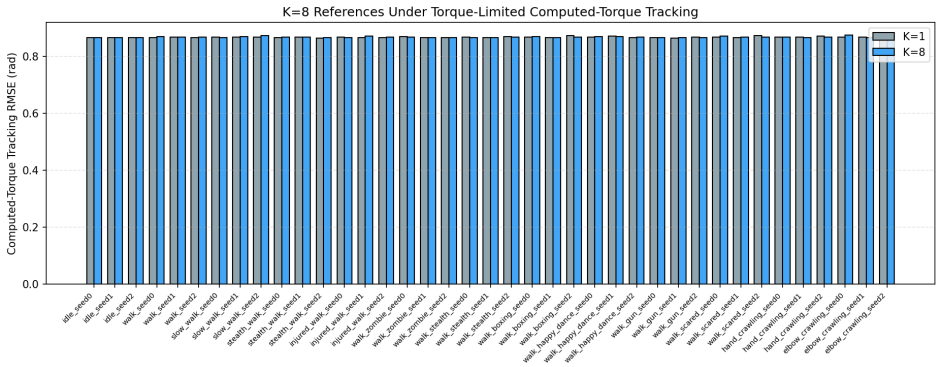

Figure 3: Controller negative control. The available torque-limited computed-torque tracker fails uniformly and does not validate the heuristic critic. This is why the paper claims screening/reranking rather than physical execution.

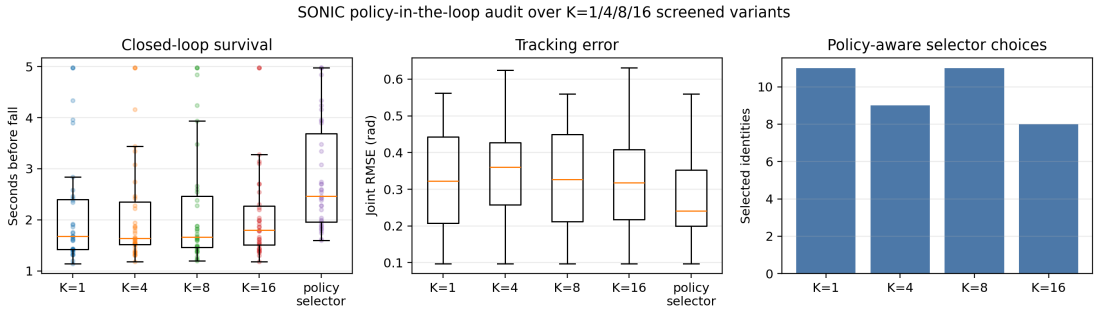

Figure 4: Corrected SONIC learned-policy audit over the 39-identity K=1/4/8/16 sweep. Inverse-dynamics-selected K values do not monotonically improve closed-loop tracking. A controller-aware selector improves short-horizon survival, but fall count remains 36/39, so this is evidence for policy-in-the-loop selection rather than solved execution.

learned SONIC harness is more relevant: after fixing the harness, K=8 has a small non-significant survival/RMSE improvement, while a policy-aware selector improves survival and RMSE significantly. Thus lower heuristic risk alone is not a validated substitute for closed-loop tracking outcomes.

5 Discussion

Best-of-K works because MotionBricks has meaningful local candidate diversity: different seeds choose different references, and Gumbel sampling changes the token trajectory toward a reference. For upright and expressive motions, some candidates have lower derivative and torque demand while preserving the broad mode label. For crawling, the candidate set remains high demand under the current robot model and critic. For idle clips, the correct policy is to skip resampling when the first candidate is already low risk.

The main limitation is execution. A publishable physical-execution claim needs a strong learned G1 tracking setup or hardware rollout in which selected references actually remain upright beyond short horizons and across dynamic motions. Our SONIC audit is a first step toward that target: the harness runs, but most non-idle clips still fall within 5s. The best next step is controller-in-the-loop selection, planner-policy co-training, or preference tuning: use controller outcomes to shift the generator distribution, rather than relying only on inverse-dynamics rejection sampling.

The prompt-control and contact audits also clarify scope. The method screens among local candidates; it does not create new semantic capability, and it does not guarantee collision-free or dynamically executable motion. Its strongest current use is as a transparent triage layer before expensive controller rollout.

6 Artifacts

Primary artifacts are:

- `results/guided_ablation_full.csv`
- `results/guided_ablation_extended.csv`
- `results/guided_extended_k1_vs_k8.png`
- `configs/prompt_suite_105.csv`
- `results/prompt_alignment.csv`
- `results/contact_quality.csv`
- `results/semantic_preservation.csv`
- `results/computed_torque_tracking.csv`
- `results/sonic_policy_mujoco_tracking_210_fixed.csv`
- `results/sonic_policy_mujoco_tracking_k_sweep_fixed.csv`
- `results/sonic_policy_multik_selector_fixed.png`
- `results/videos/comparison/*.mp4`
- `results/videos/sonic_policy_multik/*.mp4`
- `results/videos/supplementary/*.mp4`

References

[1] Jianuo Cao, Yuxin Chen, and Masayoshi Tomizuka. Claw: Composable language-annotated whole-body motion generation, 2026.

[2] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems, 2021.

[3] Chuan Guo, Shihao Zou, Xinxin Zuo, Sen Wang, Wei Ji, Xingyu Li, and Li Cheng. Generating diverse and natural 3d human motions from text. In *CVPR*, 2022.

[4] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step, 2023.

[5] Zhengyi Luo, Jinkun Cao, Alexander Winkler, Kris Kitani, and Weipeng Xu. Perpetual humanoid control for real-time simulated avatars. In *ICCV*, 2023.

[6] Xue Bin Peng, Pieter Abbeel, Sergey Levine, and Michiel van de Panne. Deepmimic: Example-guided deep reinforcement learning of physics-based character skills. *ACM Transactions on Graphics*, 2018.

[7] Xue Bin Peng, Ze Ma, Pieter Abbeel, Sergey Levine, and Angjoo Kanazawa. Amp: Adversarial motion priors for stylized physics-based character control. *ACM Transactions on Graphics*, 2021.

[8] Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling llm test-time compute optimally improves diverse reasoning tasks, 2024.

[9] Tingwu Wang, Olivier Dionne, Michael De Ruyter, David Minor, Davis Rempe, Kaifeng Zhao, Mathis Petrovich, Ye Yuan, Chenran Li, Zhengyi Luo, Brian Robison, Xavier Blackwell, Bernardo Antoniazzi, Xue Bin Peng, Yuke Zhu, and Simon Yuen. Motionbricks: Scalable real-time motions with modular latent generative model and smart primitives, 2026.

[10] Xichen Yuan, Zhe Li, Bofan Lyu, Kuangji Zuo, Yanshuo Lu, Gen Li, and Jianfei Yang. Roboforge: Physically optimized text-guided whole-body locomotion for humanoids, 2026.

[11] Jianrong Zhang, Yan Zhang, Xiaodong Cun, Yong Huang, Yong Zhang, Hongwei Zhao, Hongtao Lu, and Xi Shen. T2m-gpt: Generating human motion from textual descriptions with discrete representations. In *CVPR*, 2023.